

Measurements that can falsify the proposed mechanism attribution

Measurement	Mechanism constrained	Expected falsifiable signature
Matched-density G_mean, G50, G90	fast and two-step trajectory separation	fast path smaller after onset; two-step distribution remains finer
Densification-onset time during ramps	nucleation-limited waiting	nucleation-facile material weakens heating-rate separation
Exchange and transport relaxation	post-nucleation attainment	completion remains fast after the onset threshold
3D open/closed pore fraction	PR-prepared closed store	first step increases a persistent closed inventory
Pore D50, D90, and large-pore tail	PR/surface redistribution	schedule-dependent distribution at matched density
Connected fine-pore fraction/percolation	removable-pore topology	connectivity changes across first-step preparation
Trapped-gas pressure or accommodation proxy	finite closed accommodation	low-T2 exhaustion accompanies accommodation depletion
Grain-growth mobility versus T2	upper Chen boundary	migration rises across the upper success boundary
Interrupted first/second-step tomography	memory persistence	prepared topology persists long enough to affect G(rho)

Primary calibration targets: closed-pore fraction and accommodation trajectory